

Contents

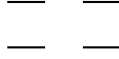
1	Orbitals And Their Symmetry	2
2	Linear Combinations: 2 Monomer- / 1 Dimer-States	3
3	Linear Combinations: 4 Monomer- / 2 Dimer-States	21
4	Linear Combinations: 8 Monomer- / 4 Dimer-States	32
5	Linear Combinations: 16 Monomer- / 8 Dimer-States	48

1 Orbitals And Their Symmetry

We use the following order of orbitals here (assuming degeneracy of LUMOs and HOMOs):

$$\begin{array}{cc} b_{1u} & \text{---} & \text{---} & a_u \\ b_{2g} & \text{---} & \text{---} & b_{3g} \end{array}$$

In the following, we will leave out the explicit labeling by orbital symmetry. Each monomer will simply be written as:



where the character of the orbital is defined by its position.

Dimer configurations will be written as their monomer occupations, by writing one monomer to the left and one to the right. The orbital order within the group of monomer orbitals follows the above mentioned definition. The transformation of

monomer orbitals into dimer orbitals is given by knowing the negative and positive linear combinations of monomer orbitals into dimer orbitals. Transforming this known equation system leads to:

$$a_u^l = +b_{1g} + a_u \tag{1a}$$

$$b_{1u}^l = +a_g + b_{1u} \tag{1b}$$

$$b_{2g}^l = +b_{3u} + b_{2g} \tag{1c}$$

$$b_{3g}^l = +b_{2u} + b_{3g} \tag{1d}$$

$$a_u^r = -b_{1g} + a_u \tag{1e}$$

$$b_{1u}^r = -a_g + b_{1u} \tag{1f}$$

$$b_{2g}^r = -b_{3u} + b_{2g} \tag{1g}$$

$$b_{3g}^r = -b_{2u} + b_{3g} \tag{1h}$$

2 Linear Combinations: 2 Monomer- / 1 Dimer-States

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & \uparrow & - & - \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} \\
& + \underbrace{(b_{1u}^l) \cdot (a_u^l) \cdot (b_{2g}^l) \cdot (b_{3g}^l)}_{l: a_g} \Rightarrow + |(b_{1u}^l) \cdot (b_{2g}^l) \cdot (b_{3g}^l) \cdot (a_u^l)| \\
& = + |(a_g + b_{1u}) \cdot (b_{3u} + b_{2g}) \cdot (b_{2u} + b_{3g}) \cdot (b_{1g} + a_u)| \\
& = + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& \quad + \left| \underbrace{a_g b_{2g} b_{2u} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2g} b_{2u} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& \quad + \left| \underbrace{b_{1u} b_{3u} b_{2u} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{3u} b_{2u} a_u}_{a_g} \right| + \left| \underbrace{b_{1u} b_{3u} b_{3g} b_{1g}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{3u} b_{3g} a_u}_{b_{1u}} \right| \\
& \quad + \left| \underbrace{b_{1u} b_{2g} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{2g} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\
& = + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& \quad + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& \quad - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\
& \quad + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \begin{pmatrix} - & - & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix} \\
& + \underbrace{(b_{1u}^r) \cdot (a_u^r) \cdot (b_{2g}^r) \cdot (b_{3g}^r)}_{r: a_g} \Rightarrow + |(b_{1u}^r) \cdot (b_{2g}^r) \cdot (b_{3g}^r) \cdot (a_u^r)| \\
& = + |(-a_g + b_{1u}) \cdot (-b_{3u} + b_{2g}) \cdot (-b_{2u} + b_{3g}) \cdot (-b_{1g} + a_u)|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{2g} b_{2u} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2g} b_{2u} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& - \left| \underbrace{b_{1u} b_{3u} b_{2u} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{3u} b_{2u} a_u}_{a_g} \right| + \left| \underbrace{b_{1u} b_{3u} b_{3g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{1u} b_{3u} b_{3g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{1u} b_{2g} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{1u} b_{2g} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) \\
& + \underbrace{(a_u^l) \cdot (b_{2g}^l)}_{\text{l: } b_{2u}} \cdot \underbrace{(a_u^r) \cdot (b_{2g}^r)}_{\text{r: } b_{2u}} \Rightarrow + |(b_{2g}^l) \cdot (b_{2g}^r) \cdot (a_u^l) \cdot (a_u^r)| \\
& = + |(+b_{3u} + b_{2g}) \cdot (-b_{3u} + b_{2g}) \cdot (+b_{1g} + a_u) \cdot (-b_{1g} + a_u)| \\
& = + \left| \underbrace{b_{3u}b_{2g}b_{1g}a_u}_{a_g} - \underbrace{b_{3u}b_{2g}a_ub_{1g}}_{a_g} - \underbrace{b_{2g}b_{3u}b_{1g}a_u}_{a_g} + \underbrace{b_{2g}b_{3u}a_ub_{1g}}_{a_g} \right| \\
& = - \left| \underbrace{b_{3u}b_{1g}b_{2g}a_u}_{a_g} - \underbrace{b_{3u}b_{1g}b_{2g}a_u}_{a_g} - \underbrace{b_{3u}b_{1g}b_{2g}a_u}_{a_g} - \underbrace{b_{3u}b_{1g}b_{2g}a_u}_{a_g} \right|
\end{aligned}$$

summarized:

$$= -4 \left| \underbrace{b_{3u}b_{1g}b_{2g}a_u}_{a_g} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \updownarrow & \uparrow & \updownarrow \end{pmatrix} \\
& + \underbrace{(a_u^l) \cdot (b_{2g}^l)}_{\text{l: } b_{2u}} \cdot \underbrace{(b_{1u}^r) \cdot (b_{2g}^r)}_{\text{r: } b_{3u}} \Rightarrow + |(b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{2g}^r) \cdot (a_u^l)| \\
& = + |(-a_g + b_{1u}) \cdot (+b_{3u} + b_{2g}) \cdot (-b_{3u} + b_{2g}) \cdot (+b_{1g} + a_u)| \\
\\
& = - \left| \underbrace{a_g b_{3u} b_{2g} b_{1g}}_{a_u} \right| - \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| + \left| \underbrace{a_g b_{2g} b_{3u} b_{1g}}_{a_u} \right| + \left| \underbrace{a_g b_{2g} b_{3u} a_u}_{b_{1g}} \right| \\
& \quad + \left| \underbrace{b_{1u} b_{3u} b_{2g} b_{1g}}_{b_{1g}} \right| + \left| \underbrace{b_{1u} b_{3u} b_{2g} a_u}_{a_u} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3u} b_{1g}}_{b_{1g}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3u} a_u}_{a_u} \right| \\
\\
& = + \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| - \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| - \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| \\
& \quad - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| - \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| - \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right|
\end{aligned}$$

summarized:

$$\begin{aligned}
& = +2 \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| - 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| - 2 \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\
\\
& + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \updownarrow & \uparrow & \updownarrow \end{pmatrix} \\
& + \underbrace{(b_{1u}^l) \cdot (b_{2g}^l)}_{\text{l: } b_{3u}} \cdot \underbrace{(a_u^r) \cdot (b_{2g}^r)}_{\text{r: } b_{2u}} \Rightarrow + |(b_{1u}^l) \cdot (b_{2g}^l) \cdot (b_{2g}^r) \cdot (a_u^r)| \\
& = + |(+a_g + b_{1u}) \cdot (+b_{3u} + b_{2g}) \cdot (-b_{3u} + b_{2g}) \cdot (-b_{1g} + a_u)| \\
\\
& = - \left| \underbrace{a_g b_{3u} b_{2g} b_{1g}}_{a_u} \right| + \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| + \left| \underbrace{a_g b_{2g} b_{3u} b_{1g}}_{a_u} \right| - \left| \underbrace{a_g b_{2g} b_{3u} a_u}_{b_{1g}} \right| \\
& \quad - \left| \underbrace{b_{1u} b_{3u} b_{2g} b_{1g}}_{b_{1g}} \right| + \left| \underbrace{b_{1u} b_{3u} b_{2g} a_u}_{a_u} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3u} b_{1g}}_{b_{1g}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3u} a_u}_{a_u} \right|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| + \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| + \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| - \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| - \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right|
\end{aligned}$$

summarized:

$$= +2 \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| + 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| - 2 \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(a_u^l) \cdot (b_{2g}^l)}_{l: b_{2u}} \cdot \underbrace{(a_u^r) \cdot (b_{3g}^r)}_{r: b_{3u}} \Rightarrow + |(b_{2g}^l) \cdot (b_{3g}^r) \cdot (a_u^l) \cdot (a_u^r)| \\
& = + |(+b_{3u} + b_{2g}) \cdot (-b_{2u} + b_{3g}) \cdot (+b_{1g} + a_u) \cdot (-b_{1g} + a_u)| \\
& = - \left| \underbrace{b_{3u}b_{2u}b_{1g}a_u}_{a_u} \right| + \left| \underbrace{b_{3u}b_{2u}a_ub_{1g}}_{a_u} \right| + \left| \underbrace{b_{3u}b_{3g}b_{1g}a_u}_{b_{1g}} \right| - \left| \underbrace{b_{3u}b_{3g}a_ub_{1g}}_{b_{1g}} \right| \\
& \quad - \left| \underbrace{b_{2g}b_{2u}b_{1g}a_u}_{b_{1g}} \right| + \left| \underbrace{b_{2g}b_{2u}a_ub_{1g}}_{b_{1g}} \right| + \left| \underbrace{b_{2g}b_{3g}b_{1g}a_u}_{a_u} \right| - \left| \underbrace{b_{2g}b_{3g}a_ub_{1g}}_{a_u} \right| \\
& = - \left| \underbrace{b_{3u}b_{2u}b_{1g}a_u}_{a_u} \right| - \left| \underbrace{b_{3u}b_{2u}b_{1g}a_u}_{a_u} \right| - \left| \underbrace{b_{3u}b_{1g}b_{3g}a_u}_{b_{1g}} \right| - \left| \underbrace{b_{3u}b_{1g}b_{3g}a_u}_{b_{1g}} \right| \\
& \quad - \left| \underbrace{b_{2u}b_{1g}b_{2g}a_u}_{b_{1g}} \right| - \left| \underbrace{b_{2u}b_{1g}b_{2g}a_u}_{b_{1g}} \right| + \left| \underbrace{b_{1g}b_{2g}b_{3g}a_u}_{a_u} \right| + \left| \underbrace{b_{1g}b_{2g}b_{3g}a_u}_{a_u} \right|
\end{aligned}$$

summarized:

$$\begin{aligned}
& = -2 \left| \underbrace{b_{3u}b_{2u}b_{1g}a_u}_{a_u} \right| - 2 \left| \underbrace{b_{3u}b_{1g}b_{3g}a_u}_{b_{1g}} \right| - 2 \left| \underbrace{b_{2u}b_{1g}b_{2g}a_u}_{b_{1g}} \right| + 2 \left| \underbrace{b_{1g}b_{2g}b_{3g}a_u}_{a_u} \right| \\
& + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(a_u^l) \cdot (b_{3g}^l)}_{l: b_{3u}} \cdot \underbrace{(a_u^r) \cdot (b_{2g}^r)}_{r: b_{2u}} \Rightarrow + |(b_{2g}^r) \cdot (b_{3g}^l) \cdot (a_u^l) \cdot (a_u^r)| \\
& = + |(-b_{3u} + b_{2g}) \cdot (+b_{2u} + b_{3g}) \cdot (+b_{1g} + a_u) \cdot (-b_{1g} + a_u)| \\
& = - \left| \underbrace{b_{3u}b_{2u}b_{1g}a_u}_{a_u} \right| + \left| \underbrace{b_{3u}b_{2u}a_ub_{1g}}_{a_u} \right| - \left| \underbrace{b_{3u}b_{3g}b_{1g}a_u}_{b_{1g}} \right| + \left| \underbrace{b_{3u}b_{3g}a_ub_{1g}}_{b_{1g}} \right| \\
& \quad + \left| \underbrace{b_{2g}b_{2u}b_{1g}a_u}_{b_{1g}} \right| - \left| \underbrace{b_{2g}b_{2u}a_ub_{1g}}_{b_{1g}} \right| + \left| \underbrace{b_{2g}b_{3g}b_{1g}a_u}_{a_u} \right| - \left| \underbrace{b_{2g}b_{3g}a_ub_{1g}}_{a_u} \right|
\end{aligned}$$

$$\begin{aligned}
= & - \left| \underbrace{b_{3u}b_{2u}b_{1g}a_u}_{a_u} \right| - \left| \underbrace{b_{3u}b_{2u}b_{1g}a_u}_{a_u} \right| + \left| \underbrace{b_{3u}b_{1g}b_{3g}a_u}_{b_{1g}} \right| + \left| \underbrace{b_{3u}b_{1g}b_{3g}a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{b_{2u}b_{1g}b_{2g}a_u}_{b_{1g}} \right| + \left| \underbrace{b_{2u}b_{1g}b_{2g}a_u}_{b_{1g}} \right| + \left| \underbrace{b_{1g}b_{2g}b_{3g}a_u}_{a_u} \right| + \left| \underbrace{b_{1g}b_{2g}b_{3g}a_u}_{a_u} \right|
\end{aligned}$$

summarized:

$$= -2 \left| \underbrace{b_{3u}b_{2u}b_{1g}a_u}_{a_u} \right| + 2 \left| \underbrace{b_{3u}b_{1g}b_{3g}a_u}_{b_{1g}} \right| + 2 \left| \underbrace{b_{2u}b_{1g}b_{2g}a_u}_{b_{1g}} \right| + 2 \left| \underbrace{b_{1g}b_{2g}b_{3g}a_u}_{a_u} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\
& + \underbrace{(a_u^l) \cdot (b_{2g}^l)}_{\text{l: } b_{2u}} \cdot \underbrace{(b_{1u}^r) \cdot (b_{3g}^r)}_{\text{r: } b_{2u}} \Rightarrow + |(b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{3g}^r) \cdot (a_u^l)| \\
& = + |(-a_g + b_{1u}) \cdot (+b_{3u} + b_{2g}) \cdot (-b_{2u} + b_{3g}) \cdot (+b_{1g} + a_u)| \\
& = + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} b_{1g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& \quad + \left| \underbrace{a_g b_{2g} b_{2u} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2g} b_{2u} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& \quad - \left| \underbrace{b_{1u} b_{3u} b_{2u} b_{1g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{3u} b_{2u} a_u}_{a_g} \right| + \left| \underbrace{b_{1u} b_{3u} b_{3g} b_{1g}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{3u} b_{3g} a_u}_{b_{1u}} \right| \\
& \quad - \left| \underbrace{b_{1u} b_{2g} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{1u} b_{2g} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\
& = + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& \quad + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& \quad + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\
& \quad - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} \\
& + \underbrace{(b_{1u}^l) \cdot (b_{3g}^l)}_{\text{l: } b_{2u}} \cdot \underbrace{(a_u^r) \cdot (b_{2g}^r)}_{\text{r: } b_{2u}} \Rightarrow + |(b_{1u}^l) \cdot (b_{2g}^r) \cdot (b_{3g}^l) \cdot (a_u^r)| \\
& = + |(+a_g + b_{1u}) \cdot (-b_{3u} + b_{2g}) \cdot (+b_{2u} + b_{3g}) \cdot (-b_{1g} + a_u)|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} b_{1g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{2g} b_{2u} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2g} b_{2u} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{1u} b_{3u} b_{2u} b_{1g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{3u} b_{2u} a_u}_{a_g} \right| + \left| \underbrace{b_{1u} b_{3u} b_{3g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{1u} b_{3u} b_{3g} a_u}_{b_{1u}} \right| \\
& - \left| \underbrace{b_{1u} b_{2g} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{2g} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\
& - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \hline \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) \\
& + \underbrace{(b_{1u}^l) \cdot (b_{2g}^l)}_{\text{l: } b_{3u}} \cdot \underbrace{(b_{1u}^r) \cdot (b_{2g}^r)}_{\text{r: } b_{3u}} \Rightarrow + |(b_{1u}^l) \cdot (b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{2g}^r)| \\
& = + |(+a_g + b_{1u}) \cdot (-a_g + b_{1u}) \cdot (+b_{3u} + b_{2g}) \cdot (-b_{3u} + b_{2g})| \\
& = + \left| \underbrace{a_g b_{1u} b_{3u} b_{2g}}_{a_g} \right| - \left| \underbrace{a_g b_{1u} b_{2g} b_{3u}}_{a_g} \right| - \left| \underbrace{b_{1u} a_g b_{3u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{1u} a_g b_{2g} b_{3u}}_{a_g} \right| \\
& = - \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right|
\end{aligned}$$

summarized:

$$= -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& + \underbrace{(b_{1u}^l) \cdot (b_{2g}^l)}_{\text{l: } b_{3u}} \cdot \underbrace{(a_u^r) \cdot (b_{3g}^r)}_{\text{r: } b_{3u}} \Rightarrow + |(b_{1u}^l) \cdot (b_{2g}^l) \cdot (b_{3g}^r) \cdot (a_u^r)| \\
& = + |(+a_g + b_{1u}) \cdot (+b_{3u} + b_{2g}) \cdot (-b_{2u} + b_{3g}) \cdot (-b_{1g} + a_u)| \\
& = + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} - \underbrace{a_g b_{3u} b_{3g} b_{1g}}_{b_{1u}} + \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& \quad + \left| \underbrace{a_g b_{2g} b_{2u} b_{1g}}_{b_{1u}} - \underbrace{a_g b_{2g} b_{2u} a_u}_{a_g} - \underbrace{a_g b_{2g} b_{3g} b_{1g}}_{a_g} + \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& \quad + \left| \underbrace{b_{1u} b_{3u} b_{2u} b_{1g}}_{b_{1u}} - \underbrace{b_{1u} b_{3u} b_{2u} a_u}_{a_g} - \underbrace{b_{1u} b_{3u} b_{3g} b_{1g}}_{a_g} + \underbrace{b_{1u} b_{3u} b_{3g} a_u}_{b_{1u}} \right| \\
& \quad + \left| \underbrace{b_{1u} b_{2g} b_{2u} b_{1g}}_{a_g} - \underbrace{b_{1u} b_{2g} b_{2u} a_u}_{b_{1u}} - \underbrace{b_{1u} b_{2g} b_{3g} b_{1g}}_{b_{1u}} + \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\
& = + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} + \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} + \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& \quad + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} + \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} - \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} + \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& \quad - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} - \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} + \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} - \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\
& \quad + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} + \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} + \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} \\
& + \underbrace{(a_u^l) \cdot (b_{3g}^l)}_{\text{l: } b_{3u}} \cdot \underbrace{(b_{1u}^r) \cdot (b_{2g}^r)}_{\text{r: } b_{3u}} \Rightarrow + |(b_{1u}^r) \cdot (b_{2g}^r) \cdot (b_{3g}^l) \cdot (a_u^l)| \\
& = + |(-a_g + b_{1u}) \cdot (-b_{3u} + b_{2g}) \cdot (+b_{2u} + b_{3g}) \cdot (+b_{1g} + a_u)|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{2g} b_{2u} b_{1g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2g} b_{2u} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& - \left| \underbrace{b_{1u} b_{3u} b_{2u} b_{1g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{3u} b_{2u} a_u}_{a_g} \right| - \left| \underbrace{b_{1u} b_{3u} b_{3g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{1u} b_{3u} b_{3g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{1u} b_{2g} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{2g} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} b_{1g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& + \underbrace{(b_{1u}^l) \cdot (b_{2g}^l)}_{\text{l: } b_{3u}} \cdot \underbrace{(b_{1u}^r) \cdot (b_{3g}^r)}_{\text{r: } b_{2u}} \Rightarrow + |(b_{1u}^l) \cdot (b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{3g}^r)| \\
& = + |(+a_g + b_{1u}) \cdot (-a_g + b_{1u}) \cdot (+b_{3u} + b_{2g}) \cdot (-b_{2u} + b_{3g})| \\
& = - \left| \underbrace{a_g b_{1u} b_{3u} b_{2u}}_{a_u} \right| + \left| \underbrace{a_g b_{1u} b_{3u} b_{3g}}_{b_{1g}} \right| - \left| \underbrace{a_g b_{1u} b_{2g} b_{2u}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| \\
& \quad + \left| \underbrace{b_{1u} a_g b_{3u} b_{2u}}_{a_u} \right| - \left| \underbrace{b_{1u} a_g b_{3u} b_{3g}}_{b_{1g}} \right| + \left| \underbrace{b_{1u} a_g b_{2g} b_{2u}}_{b_{1g}} \right| - \left| \underbrace{b_{1u} a_g b_{2g} b_{3g}}_{a_u} \right| \\
& = - \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| - \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| - \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| \\
& \quad - \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| - \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| - \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right|
\end{aligned}$$

summarized:

$$= -2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| - 2 \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| - 2 \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| + 2 \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} \\
& + \underbrace{(b_{1u}^l) \cdot (b_{3g}^l)}_{\text{l: } b_{2u}} \cdot \underbrace{(b_{1u}^r) \cdot (b_{2g}^r)}_{\text{r: } b_{3u}} \Rightarrow + |(b_{1u}^l) \cdot (b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{3g}^l)| \\
& = + |(+a_g + b_{1u}) \cdot (-a_g + b_{1u}) \cdot (-b_{3u} + b_{2g}) \cdot (+b_{2u} + b_{3g})| \\
& = - \left| \underbrace{a_g b_{1u} b_{3u} b_{2u}}_{a_u} \right| - \left| \underbrace{a_g b_{1u} b_{3u} b_{3g}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{1u} b_{2g} b_{2u}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| \\
& \quad + \left| \underbrace{b_{1u} a_g b_{3u} b_{2u}}_{a_u} \right| + \left| \underbrace{b_{1u} a_g b_{3u} b_{3g}}_{b_{1g}} \right| - \left| \underbrace{b_{1u} a_g b_{2g} b_{2u}}_{b_{1g}} \right| - \left| \underbrace{b_{1u} a_g b_{2g} b_{3g}}_{a_u} \right|
\end{aligned}$$

$$\begin{aligned}
= & - \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right|
\end{aligned}$$

summarized:

$$= -2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + 2 \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| + 2 \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| + 2 \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) \\
& + \underbrace{(a_u^l) \cdot (b_{3g}^l)}_{\text{l: } b_{3u}} \cdot \underbrace{(a_u^r) \cdot (b_{3g}^r)}_{\text{r: } b_{3u}} \Rightarrow + |(b_{3g}^l) \cdot (b_{3g}^r) \cdot (a_u^l) \cdot (a_u^r)| \\
& = + |(+b_{2u} + b_{3g}) \cdot (-b_{2u} + b_{3g}) \cdot (+b_{1g} + a_u) \cdot (-b_{1g} + a_u)| \\
& = + \left| \underbrace{b_{2u} b_{3g} b_{1g} a_u}_{a_g} \right| - \left| \underbrace{b_{2u} b_{3g} a_u b_{1g}}_{a_g} \right| - \left| \underbrace{b_{3g} b_{2u} b_{1g} a_u}_{a_g} \right| + \left| \underbrace{b_{3g} b_{2u} a_u b_{1g}}_{a_g} \right| \\
& = - \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

summarized:

$$= -4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& + \underbrace{(a_u^l) \cdot (b_{3g}^l)}_{\text{l: } b_{3u}} \cdot \underbrace{(b_{1u}^r) \cdot (b_{3g}^r)}_{\text{r: } b_{2u}} \Rightarrow + |(b_{1u}^r) \cdot (b_{3g}^l) \cdot (b_{3g}^r) \cdot (a_u^l)| \\
& = + |(-a_g + b_{1u}) \cdot (+b_{2u} + b_{3g}) \cdot (-b_{2u} + b_{3g}) \cdot (+b_{1g} + a_u)| \\
& = - \left| \underbrace{a_g b_{2u} b_{3g} b_{1g}}_{a_u} \right| - \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| + \left| \underbrace{a_g b_{3g} b_{2u} b_{1g}}_{a_u} \right| + \left| \underbrace{a_g b_{3g} b_{2u} a_u}_{b_{1g}} \right| \\
& \quad + \left| \underbrace{b_{1u} b_{2u} b_{3g} b_{1g}}_{b_{1g}} \right| + \left| \underbrace{b_{1u} b_{2u} b_{3g} a_u}_{a_u} \right| - \left| \underbrace{b_{1u} b_{3g} b_{2u} b_{1g}}_{b_{1g}} \right| - \left| \underbrace{b_{1u} b_{3g} b_{2u} a_u}_{a_u} \right| \\
& = + \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| + \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| \\
& \quad - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| - \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| - \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$

summarized:

$$= +2 \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - 2 \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| - 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| - 2 \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& + \underbrace{(b_{1u}^l) \cdot (b_{3g}^l)}_{\text{l: } b_{2u}} \cdot \underbrace{(a_u^r) \cdot (b_{3g}^r)}_{\text{r: } b_{3u}} \Rightarrow + |(b_{1u}^l) \cdot (b_{3g}^l) \cdot (b_{3g}^r) \cdot (a_u^r)| \\
& = + |(+a_g + b_{1u}) \cdot (+b_{2u} + b_{3g}) \cdot (-b_{2u} + b_{3g}) \cdot (-b_{1g} + a_u)| \\
& = - \left| \underbrace{a_g b_{2u} b_{3g} b_{1g}}_{a_u} \right| + \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| + \left| \underbrace{a_g b_{3g} b_{2u} b_{1g}}_{a_u} \right| - \left| \underbrace{a_g b_{3g} b_{2u} a_u}_{b_{1g}} \right| \\
& \quad - \left| \underbrace{b_{1u} b_{2u} b_{3g} b_{1g}}_{b_{1g}} \right| + \left| \underbrace{b_{1u} b_{2u} b_{3g} a_u}_{a_u} \right| + \left| \underbrace{b_{1u} b_{3g} b_{2u} b_{1g}}_{b_{1g}} \right| - \left| \underbrace{b_{1u} b_{3g} b_{2u} a_u}_{a_u} \right|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| + \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| + \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| + \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| - \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| - \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$

summarized:

$$= +2 \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| + 2 \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| + 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| - 2 \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(b_{1u}^l) \cdot (b_{3g}^l)}_{\text{l: } b_{2u}} \cdot \underbrace{(b_{1u}^r) \cdot (b_{3g}^r)}_{\text{r: } b_{2u}} \Rightarrow + |(b_{1u}^l) \cdot (b_{1u}^r) \cdot (b_{3g}^l) \cdot (b_{3g}^r)| \\
& = + |(+a_g + b_{1u}) \cdot (-a_g + b_{1u}) \cdot (+b_{2u} + b_{3g}) \cdot (-b_{2u} + b_{3g})| \\
& = + \left| \underbrace{a_g b_{1u} b_{2u} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{1u} b_{3g} b_{2u}}_{a_g} \right| - \left| \underbrace{b_{1u} a_g b_{2u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{1u} a_g b_{3g} b_{2u}}_{a_g} \right| \\
& = - \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right|
\end{aligned}$$

summarized:

$$= -4 \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right|$$

3 Linear Combinations: 4 Monomer- / 2 Dimer-States

$$+ \begin{pmatrix} \uparrow & \uparrow & - & - \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & - & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_g

$$= + (|(b_{1u}^l) \cdot (b_{2g}^l) \cdot (b_{3g}^l) \cdot (a_u^l)|) + (|(b_{1u}^r) \cdot (b_{2g}^r) \cdot (b_{3g}^r) \cdot (a_u^r)|)$$

$$= +2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right|$$

$$+ 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|$$

$$+ \begin{pmatrix} \uparrow & \uparrow & - & - \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & - & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_g

$$= + (|(b_{1u}^l) \cdot (b_{2g}^l) \cdot (b_{3g}^l) \cdot (a_u^l)|) - (|(b_{1u}^r) \cdot (b_{2g}^r) \cdot (b_{3g}^r) \cdot (a_u^r)|)$$

$$= +2 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right|$$

$$- 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1u}} \right| - 2 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|$$

$$+ \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \uparrow \end{array} \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \uparrow \end{array} \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \uparrow \end{array} \right)$$

enthaltene Monomer-Symmetrien = b_{2u}

$$= \qquad \qquad \qquad + (|(b_{2g}^l) \cdot (b_{2g}^r) \cdot (a_u^l) \cdot (a_u^r)|) \qquad \qquad + \qquad \qquad (|(b_{2g}^l) \cdot (b_{2g}^r) \cdot (a_u^l) \cdot (a_u^r)|)$$

$$= \qquad \qquad \qquad -8 \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \uparrow & \text{---} \\ \uparrow & \updownarrow & \uparrow & \updownarrow \end{pmatrix} + \begin{pmatrix} \uparrow & \text{---} & \text{---} & \uparrow \\ \uparrow & \updownarrow & \uparrow & \updownarrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_{3u}, b_{2u}

$$= \qquad \qquad \qquad + (|(b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{2g}^r) \cdot (a_u^l)|) \qquad \qquad + \qquad \qquad (|(b_{1u}^l) \cdot (b_{2g}^l) \cdot (b_{2g}^r) \cdot (a_u^r)|)$$

$$= \qquad \qquad \qquad + 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| \qquad \qquad - \qquad \qquad 4 \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \uparrow & \text{---} \\ \uparrow & \updownarrow & \uparrow & \updownarrow \end{pmatrix} - \begin{pmatrix} \uparrow & \text{---} & \text{---} & \uparrow \\ \uparrow & \updownarrow & \uparrow & \updownarrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_{3u}, b_{2u}

$$= \qquad \qquad \qquad + (|(b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{2g}^r) \cdot (a_u^l)|) \qquad \qquad - \qquad \qquad (|(b_{1u}^l) \cdot (b_{2g}^l) \cdot (b_{2g}^r) \cdot (a_u^r)|)$$

$$= \qquad \qquad \qquad - 4 \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| \qquad \qquad - \qquad \qquad 4 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \text{---} & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \text{---} & \uparrow & \text{---} & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_{3u}, b_{2u}

$$= + (|(b_{2g}^l) \cdot (b_{3g}^r) \cdot (a_u^l) \cdot (a_u^r)|) + (|(b_{2g}^r) \cdot (b_{3g}^l) \cdot (a_u^l) \cdot (a_u^r)|)$$

$$= -4 \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| + 4 \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \text{---} & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \text{---} & \uparrow & \text{---} & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_{3u}, b_{2u}

$$= + (|(b_{2g}^l) \cdot (b_{3g}^r) \cdot (a_u^l) \cdot (a_u^r)|) - (|(b_{2g}^r) \cdot (b_{3g}^l) \cdot (a_u^l) \cdot (a_u^r)|)$$

$$= -4 \left| \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} \right|$$

$$+ \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_{2u}

$$= + (|(b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{3g}^r) \cdot (a_u^l)|) + (|(b_{1u}^l) \cdot (b_{2g}^r) \cdot (b_{3g}^l) \cdot (a_u^r)|)$$

$$= +2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| \\ - 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|$$

$$+ \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_{2u}

$$= + (|(b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{3g}^r) \cdot (a_u^l)|) - (|(b_{1u}^l) \cdot (b_{2g}^r) \cdot (b_{3g}^l) \cdot (a_u^r)|)$$

$$= +2 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - 2 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + 2 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 2 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|$$

$$+ \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \right)$$

enthaltene Monomer-Symmetrien = b_{3u}

$$= \qquad \qquad \qquad + (|(b_{1u}^l) \cdot (b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{2g}^r)|) \qquad \qquad + \qquad \qquad (|(b_{1u}^l) \cdot (b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{2g}^r)|)$$

$$= \qquad \qquad \qquad -8 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right|$$

$$+ \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_{3u}

$$= + (|(b_{1u}^l) \cdot (b_{2g}^l) \cdot (b_{3g}^r) \cdot (a_u^r)|) + (|(b_{1u}^r) \cdot (b_{2g}^r) \cdot (b_{3g}^l) \cdot (a_u^l)|)$$

$$= +2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| \\ - 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|$$

$$+ \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_{3u}

$$= + (|(b_{1u}^l) \cdot (b_{2g}^l) \cdot (b_{3g}^r) \cdot (a_u^r)|) - (|(b_{1u}^r) \cdot (b_{2g}^r) \cdot (b_{3g}^l) \cdot (a_u^l)|)$$

$$= -2 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - 2 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 2 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|$$

$$+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right)$$

enthaltene Monomer-Symmetrien = b_{3u}, b_{2u}

$$= \quad \quad \quad + (|(b_{1u}^l) \cdot (b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{3g}^r)|) \quad \quad + \quad \quad (|(b_{1u}^l) \cdot (b_{1u}^r) \cdot (b_{2g}^r) \cdot (b_{3g}^l)|)$$

$$= \quad \quad \quad -4 \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| \quad \quad + \quad \quad 4 \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right|$$

$$+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right)$$

enthaltene Monomer-Symmetrien = b_{3u}, b_{2u}

$$= \quad \quad \quad + (|(b_{1u}^l) \cdot (b_{1u}^r) \cdot (b_{2g}^l) \cdot (b_{3g}^r)|) \quad \quad - \quad \quad (|(b_{1u}^l) \cdot (b_{1u}^r) \cdot (b_{2g}^r) \cdot (b_{3g}^l)|)$$

$$= \quad \quad \quad -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| \quad \quad - \quad \quad 4 \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right|$$

enthaltene Monomer-Symmetrien = b_{3u}

$$+ \left(|(b_{3g}^l) \cdot (b_{3g}^r) \cdot (a_u^l) \cdot (a_u^r)| \right) + \left(|(b_{3g}^l) \cdot (b_{3g}^r) \cdot (a_u^l) \cdot (a_u^r)| \right)$$

$$= -8 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \uparrow & \text{---} \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & \text{---} & \text{---} & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_{3u}, b_{2u}

$$= + (|(b_{1u}^r) \cdot (b_{3g}^l) \cdot (b_{3g}^r) \cdot (a_u^l)|) + (|(b_{1u}^l) \cdot (b_{3g}^l) \cdot (b_{3g}^r) \cdot (a_u^r)|)$$

$$= +4 \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - 4 \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \uparrow & \text{---} \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & \text{---} & \text{---} & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_{3u}, b_{2u}

$$= + (|(b_{1u}^r) \cdot (b_{3g}^l) \cdot (b_{3g}^r) \cdot (a_u^l)|) - (|(b_{1u}^l) \cdot (b_{3g}^l) \cdot (b_{3g}^r) \cdot (a_u^r)|)$$

$$= -4 \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right|$$

enthaltene Monomer-Symmetrien = b_{2u}

$$+ \left(|(b_{1u}^l) \cdot (b_{1u}^r) \cdot (b_{3g}^l) \cdot (b_{3g}^r)| \right) \quad + \quad \left(|(b_{1u}^l) \cdot (b_{1u}^r) \cdot (b_{3g}^l) \cdot (b_{3g}^r)| \right)$$

$$= -8 \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right|$$

4 Linear Combinations: 8 Monomer- / 4 Dimer-States

$i^3 b_{2u} * i^3 b_{2u}$:

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
& \quad + 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& \quad - 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$i^3 b_{2u} * e^3 b_{2u}:$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& + \left(\begin{array}{c} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(+4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| \\
& \quad + 2 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 2 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\
& \quad + 2 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$i^3 b_{2u} * e^3 b_{3u}:$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
& \quad + 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& \quad - 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$i^3 b_{2u} * i^3 b_{3u}:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\ - \left(\begin{array}{cccc} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| & - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| & - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| & - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| & + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| & + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| & + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| & - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| & - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| & - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| & + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| & - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\ + \left(\begin{array}{cccc} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| & - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| & - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| & + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| & - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| & + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| & - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| & - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| & + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| & + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\ - \left(+4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right)$$

$$= -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| \\ + 2 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 2 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\ + 2 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$e^3 b_{2u} * i^3 b_{2u}:$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
& + \left(\begin{array}{c} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(+4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - 2 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| \\
& \quad - 2 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + 2 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\
& \quad - 2 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$e^3 b_{2u} * e^3 b_{2u}:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\ + \left(\begin{array}{l} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\ + \left(\begin{array}{l} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\ + \left(-4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right)$$

$$= -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\ - 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\ + 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$e^3 b_{2u} * e^3 b_{3u}:$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
& + \left(\begin{array}{c} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(+4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - 2 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| \\
& \quad - 2 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + 2 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\
& \quad - 2 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$e^3 b_{2u} * i^3 b_{3u}:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\ + \left(\begin{array}{l} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\ + \left(\begin{array}{l} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\ + \left(-4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right)$$

$$= -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\ - 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\ + 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$e^3 b_{3u} * i^3 b_{2u}:$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
& \quad + 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& \quad - 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$e^3 b_{3u} * e^3 b_{2u}:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) + \left(\begin{array}{c} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) - \left(+4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right)$$

$$= -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| \\ + 2 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 2 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\ + 2 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$e^3 b_{3u} * e^3 b_{3u}:$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
& \quad + 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& \quad - 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$e^3 b_{3u} * i^3 b_{3u}:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\ - \left(\begin{array}{cccc} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| & - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| & - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| & - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| & + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| & + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| & + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| & - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| & - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| & - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| & + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| & - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\ + \left(\begin{array}{cccc} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| & - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| & - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| & + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| & - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| & + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| & - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| & - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| & + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| & + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\ - \left(+4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right)$$

$$= -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 2 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| \\ + 2 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 2 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\ + 2 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$i^3 b_{3u} * i^3 b_{2u}:$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
& + \left(\begin{array}{c} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(+4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - 2 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| \\
& \quad - 2 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + 2 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\
& \quad - 2 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$i^3 b_{3u} * e^3 b_{2u}:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\ + \left(\begin{array}{l} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\ + \left(\begin{array}{l} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\ + \left(-4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right)$$

$$= -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\ - 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\ + 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$i^3 b_{3u} * e^3 b_{3u}:$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
& + \left(\begin{array}{c} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(+4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 2 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - 2 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| \\
& \quad - 2 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + 2 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + 2 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\
& \quad - 2 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$i^3 b_{3u} * i^3 b_{3u}:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\ + \left(\begin{array}{l} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\ + \left(\begin{array}{l} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\ + \left(-4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right)$$

$$= -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\ - 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\ + 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

5 Linear Combinations: 16 Monomer- / 8 Dimer-States

identical $i^3 b_{2u}$ and $i^3 b_{2u}$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& + \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
& \quad + 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& \quad - 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \begin{array}{cc} \uparrow & - \\ \downarrow & \downarrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \right) \\
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \begin{array}{cc} \uparrow & - \\ \downarrow & \downarrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \right)
\end{aligned}$$

[illegible]

$$= -8 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 8 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\
& - \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right)
\end{aligned}$$

[illegible]

$$\begin{aligned}
&= -4 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + 4 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
&\quad - 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 4 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 4 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + 4 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|
\end{aligned}$$

$$\begin{aligned}
&+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
&+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}
\end{aligned}$$

[illegible]

$$\begin{aligned}
&= -8 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 4 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 4 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
&\quad + 4 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 4 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
&\quad - 4 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 8 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$\begin{aligned}
&+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) \\
&- \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right)
\end{aligned}$$

[illegible]

$$= 0 \quad \Rightarrow a_g$$

$$\begin{aligned}
 &+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
 &+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}
 \end{aligned}$$

[illegible]

$$= -8 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 8 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\
& - \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right)
\end{aligned}$$

[illegible]

$$\begin{aligned}
&= -4 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + 4 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
&\quad - 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 4 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 4 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + 4 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|
\end{aligned}$$

identical $e^3 b_{2u}$ and $e^3 b_{2u}$

$$\begin{aligned}
&+ \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) \\
&\quad + \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
&\quad + \left(\begin{array}{cccc} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| & + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| & - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| & - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| & - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| & - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| & + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| & + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| & + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| & - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| & + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
&\quad + \left(\begin{array}{cccc} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| & - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| & + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| & - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| & + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| & - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| & + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| & - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| & - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| & + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| & + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
&\quad + \left(-4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right)
\end{aligned}$$

$$\begin{aligned}
&= -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
&\quad - 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
&\quad + 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$\begin{aligned}
&+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) \\
&+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right)
\end{aligned}$$

[illegible]

$$= -8 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 8 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\
& - \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right)
\end{aligned}$$

[illegible]

$$\begin{aligned}
&= +4 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - 4 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
&\quad + 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + 4 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + 4 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - 4 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|
\end{aligned}$$

$$\begin{aligned}
&+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
&+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}
\end{aligned}$$

[illegible]

$$\begin{aligned}
&= -8 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 4 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 4 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 4 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
&\quad - 4 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 4 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 4 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 4 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
&\quad + 4 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 8 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$\begin{aligned}
&+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) \\
&- \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right)
\end{aligned}$$

[illegible]

$$= 0 \Rightarrow a_g$$

identical $e^3 b_{3u}$ and $e^3 b_{3u}$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
& \quad + 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& \quad - 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \right) \\
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \right)
\end{aligned}$$

[illegible]

$$= -8 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 8 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\
& - \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right)
\end{aligned}$$

[illegible]

$$\begin{aligned}
&= -4 \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + 4 \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
&\quad - 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 4 \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 4 \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + 4 \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|
\end{aligned}$$

identical $i^3 b_{3u}$ and $i^3 b_{3u}$

$$\begin{aligned}
&+ \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow\uparrow & \uparrow \end{array} \right) \\
&+ \left(-4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \right) \\
&+ \left(\begin{array}{c} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
&+ \left(\begin{array}{c} + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{array} \right) \\
&+ \left(-4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \right)
\end{aligned}$$

$$\begin{aligned}
&= -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
&\quad - 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
&\quad + 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$
